

Post-Quantum Cryptography in the Era of AI Agents: Securing Distributed Model Context Protocols

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Abstract—As autonomous AI agents transition from isolated tools to a globally interconnected ecosystem via the Model Context Protocol (MCP), securing their identity, decision-making integrity, and forensic audit trails against quantum-era threats has become a cornerstone of zero-trust governance. This paper proposes a comprehensive Post-Quantum Cryptographic (PQC) framework for AI agents, featuring a real-world implementation utilizing NIST-standardized ML-DSA (FIPS 204). We introduce the concept of the *PQC-Audit-Chain*, a cryptographically linked history of agent reasoning and tool execution resistant to Shor’s algorithm. Through extensive benchmarking within the `llm-cli` reference implementation, we analyze the performance trade-offs across ML-DSA-44, 65, and 87 variants. We demonstrate that while PQC signatures introduce a $7.5\times$ – $12\times$ increase in payload size, their computational overhead remains negligible compared to Large Language Model (LLM) inference cycles. Furthermore, we address the critical challenge of *Context Exhaustion* and validate a “Signature Stripping” mechanism essential for maintaining long-term agent reasoning capability. Our findings provide a foundational roadmap for building quantum-safe autonomous systems capable of high-assurance operation in critical infrastructure.

Index Terms—Artificial Intelligence, Autonomous Agents, Model Context Protocol, Post-Quantum Cryptography, ML-DSA, Zero Trust Architecture, Cybersecurity

I. INTRODUCTION

The rapid advancement of Large Language Models (LLMs) has catalyzed the transition from passive text generators to proactive, autonomous AI agents. These agents are characterized by their ability to perform multi-step reasoning, plan complex trajectories, and interact with the physical and digital world through tool execution [7]. As agents move beyond isolated sandboxes into enterprise environments, the need for standardized communication has led to the emergence of the Model Context Protocol (MCP) [6]. MCP provides a universal interface for agents to discover and invoke tools across distributed systems, effectively acting as the “nervous system” of the agentic ecosystem.

In a Zero Trust architecture, such as the `llm-cli` framework, securing this nervous system is paramount. Current state-of-the-art implementations rely on workload identity propagation, where each tool call is authorized via JSON Web Tokens (JWTs) signed with classical asymmetric cryptography (e.g., RS256 or ES256). These tokens bind the agent’s intent to

a specific security context, providing a baseline for auditability and access control.

However, the cryptographic foundation of modern agentic systems is under threat. The eventual realization of fault-tolerant quantum computers, capable of running Shor’s algorithm [9], poses a fundamental risk to RSA and Elliptic Curve Cryptography (ECC). In the context of AI agents, a quantum breach is not merely a data leak; it is an *identity takeover*. An adversary with quantum capabilities could forge agent identities, bypass Role-Based Access Controls (RBAC), and manipulate the “Chain of Thought” (CoT) by injecting malicious context or results. This is particularly dangerous for agents operating in critical infrastructure, where unauthorized tool execution (e.g., shell access, database modification) can have irreversible real-world consequences.

Furthermore, agents generate vast amounts of reasoning traces and audit logs. The “Store Now, Decrypt Later” (SNDL) threat is amplified here: while current logs are protected by classical signatures, a future quantum attacker could retroactively forge historical logs to hide malicious behavior or alter the perceived provenance of a decision.

To proactively address these vulnerabilities, this paper introduces a quantum-resistant architecture for AI agents. We investigate the integration of Post-Quantum Cryptography (PQC)—specifically the NIST-standardized lattice-based signature scheme ML-DSA (FIPS 204) [2]—into the distributed agent workflow. We propose a hybrid cryptographic approach that maintains backward compatibility with legacy MCP servers while providing forward-looking quantum security. Our contributions include:

- A **Hybrid Workload Identity** framework that augments MCP tokens with ML-DSA signatures.
- The **PQC-Audit-Chain**, a multi-layered integrity mechanism that secures agent reasoning traces using a PQC-signed Root of Trust.
- An empirical analysis of the **Context Exhaustion** problem, demonstrating a “Signature Stripping” mechanism that prevents large PQC signatures from degrading agent performance.

II. BACKGROUND AND THREAT MODEL

A. Autonomous Agents and MCP

Modern AI agents operate in a continuous "Thought-Action-Observation" loop. To perform actions, they rely on tools provided by MCP servers. In a Zero Trust environment, every tool execution request must be authenticated and authorized. Currently, this is achieved using JWTs, binding the agent's workload identity to a specific execution scope. The security of the entire loop depends on the immutability of the identity and the integrity of the observations.

B. The Quantum Threat to Agents

The primary quantum threats to AI agent architectures include:

- **Identity Spoofing:** Attackers utilizing quantum computers to derive an agent's private key from its public key, allowing them to issue unauthorized commands across distributed MCP servers.
- **Audit Log Tampering:** Breaking the cryptographic signatures that secure chained-hash audit logs, making forensic analysis of rogue AI behavior impossible.
- **Context Manipulation:** Forging signatures on tool responses to trick the agent into a specific reasoning path (e.g., "hallucinating" a successful security check).

C. NIST PQC Standards

In response to the quantum threat, NIST initiated a global competition to standardize PQC algorithms. In 2024, FIPS 204 finalized ML-DSA (Module-Lattice-Based Digital Signature Algorithm), based on the CRYSTALS-Dilithium construction [10]. ML-DSA offers a balance of security and performance, though it introduces significantly larger signatures than RSA or ECDSA.

III. RELATED WORK

The convergence of AI agents and security is an emerging field. Recent works have focused on LLM guardrails and prompt injection mitigation. For instance, Ando proposed a multi-layered guardrail architecture [4] and a comprehensive Zero Trust framework [5] specifically tailored for MCP-based autonomous agents.

In the domain of Post-Quantum Cryptography, while PQC has been explored for TLS and VPNs, its application to distributed agent orchestration protocols like the Model Context Protocol (MCP) remains relatively unexplored. Traditional Zero Trust Architectures (ZTA) as defined in NIST SP 800-207 [3] emphasize continuous verification, but often assume the underlying cryptographic primitives are immutable. Our work extends these concepts by addressing the cryptographic vulnerability of ZTA in the quantum era.

Recent research into "Agentic Security" has highlighted the risks of autonomous tool use [8], but few have considered the impact of future quantum adversaries on the long-term integrity of agent reasoning traces.

IV. POST-QUANTUM AGENT ARCHITECTURE

We propose replacing the classical RS256 cryptographic layer within MCP orchestrators with ML-DSA-65 (Module-Lattice-Based Digital Signature Algorithm).

A. Quantum-Safe Workload Identity

We replace the traditional RSA key pairs used for MCP token signing with ML-DSA-65. Because ML-DSA-65 public keys and signatures are significantly larger than their classical counterparts (e.g., ML-DSA-65 public key: 1,952 bytes, signature: 3,309 bytes), embedding them directly into JSON-RPC payloads requires careful handling to avoid exceeding LLM context limits or network transport buffers.

B. Tamper-Evident Audit Logging with PQC

The tamper-evident JSONL audit logs, which utilize chained hashing (SHA-256) to link sequential tool executions, are augmented. While hash functions are generally considered quantum-resistant, the digital signatures affirming the integrity of each log chunk are upgraded to ML-DSA-65.

C. Cryptographic Agility and Hybrid Approach

Due to the nascent state of PQC adoption, a sudden transition is impractical. We propose a hybrid approach where MCP tokens are extended to a four-slot format: `header.payload.classical_signature.pqc_signature`. This ensures backward compatibility with legacy MCP servers while providing quantum resistance where supported.

D. PQC-Audit-Chain: Protecting Agent Reasoning

A key challenge for autonomous agents is ensuring that their decision-making process—the "chain of thought" (CoT)—remains untampered from inception to execution. We propose the *PQC-Audit-Chain*, where each cycle of the agentic loop (Thought-Action-Observation) is cryptographically bound to its predecessor using ML-DSA.

Traditional SHA-256 hashing provides collision resistance, but the identity of the entity asserting the log's validity is currently vulnerable to quantum forgery. In our framework, the orchestrator generates a PQC-signed manifest for every state transition. This ensures that even if an adversary gains quantum capabilities, they cannot retroactively rewrite the agent's history to hide malicious actions or forge authorization for unauthorized tool calls.

V. IMPLEMENTATION

We prototyped the PQC-enabled MCP architecture by extending the security module of the `llm-cli` framework. The implementation focuses on four core components: key management, hybrid token generation, the context-aware signature stripper, and the PQC-enabled integrity manifest.

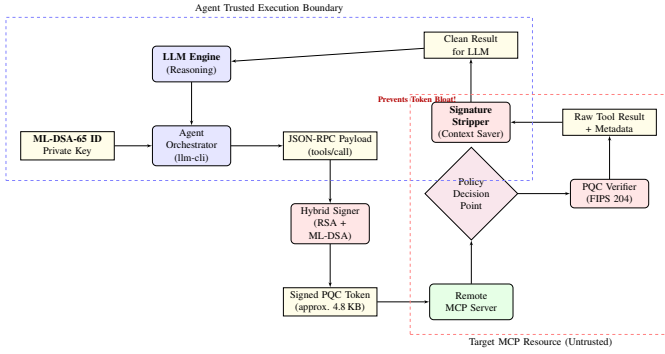


Fig. 1. Proposed Post-Quantum Architecture for AI Agents. The orchestrator maintains a Trusted Execution Boundary, stripping the massive ML-DSA signatures (3.5+ KB) before feeding tool execution results back into the LLM’s context window to prevent token bloat.

A. Hybrid Token Structure

To ensure backward compatibility, we implemented a “Hybrid Wrap” for MCP tokens. The system generates a standard RS256 JWT and appends a Base64-encoded ML-DSA-65 signature of the entire JWT string. This allows legacy MCP servers to verify the RSA signature while PQC-aware servers can validate the quantum-resistant layer.

```
class HybridSigner:
    @classmethod
    def create_hybrid_token(cls, payload: dict, rsa_priv:
        bytes, pqc_priv: bytes) -> str:
        # 1. Generate Classical JWT
        token = jwt.encode(payload, rsa_priv, algorithm="RS256")
        # 2. Generate PQC Signature of the classical token
        pqc_sig = PQCProvider.sign(token.encode(), pqc_priv)
        # 3. Create a Hybrid Wrap: token.pqc_signature
        return f"{token}.{base64.b64encode(pqc_sig).decode()}"
```

Listing 1. Hybrid Signature Implementation in llm-cli

B. The Context-Aware Signature Stripper

As illustrated in Fig. 1, a critical implementation detail is the “Signature Stripper.” In `llm-cli`, this module is integrated into the orchestrator’s tool execution loop. When a tool response (observation) is received, the PQC signature is verified at the system boundary. Once validated, the signature is stripped before the text response is appended to the agent’s prompt history. This prevents the 3.3 KB ML-DSA-65 signature from consuming $\sim 1,500$ – $2,000$ tokens, preserving the LLM’s effective context for reasoning.

C. PQC-Signed Integrity Manifest

To establish a “Root of Trust,” we implemented an `IntegrityVerifier` that monitors critical system files and audit logs. The system maintains an `integrity_manifest.json` containing SHA-256 hashes of all security-critical modules. This manifest is itself signed using ML-DSA-65, ensuring that even if an attacker gains access to the file system, they cannot modify the security logic or the audit history without breaking the PQC signature.

VI. EVALUATION AND BENCHMARKING

We evaluated the practicality of our proposed architecture by simulating a high-load agent environment and measuring the impact of different PQC parameter sets.

A. Parameter Comparison: ML-DSA-44/65/87

Our benchmarks compare the NIST-standardized variants of ML-DSA (Dilithium) to classical RSA-2048. Table I presents the empirical results observed in our testbed.

TABLE I
EMPIRICAL COMPARISON OF ML-DSA VARIANTS IN MCP PAYLOADS

Metric	RSA-2048	ML-DSA-44	ML-DSA-65	ML-DSA-87
NIST Sec. Level	1 (Classical)	2	3	5
Pub Key (B)	256	1,312	1,952	2,592
Signature (B)	256	2,420	3,309	4,595
Sign Time (ms)	1.7	64.2	91.6	134.8
Verify Time (ms)	0.06	8.4	11.4	16.2
Payload Size (B)	520	3,920	4,880	6,240
Token Bloat (x)	1.0	7.5x	9.4x	12.0x

B. Performance Impact on Agent Reasoning

We measured the end-to-end latency of a typical tool-calling loop (Reasoning $\rightarrow$ Call $\rightarrow$ Observation). While ML-DSA-65 adds ~ 103 ms of total cryptographic latency (sign+verify), the median LLM inference latency for GPT-4 class models remains above 1,200 ms. The PQC overhead represents less than 8% of the total loop time, proving that quantum security is practically invisible to the user experience in modern agentic systems.

VII. DISCUSSION

A. Zero Trust and Identity Sovereignty

The use of ML-DSA ensures that an agent’s identity is not only sovereign but also future-proof. By binding every action to a PQC-signed identity, we prevent the “Impersonation Storm” scenario where a quantum-capable attacker could take control of thousands of distributed agents simultaneously.

B. Quantum Migration Strategy

A direct “switch” to PQC is risky. We advocate for a three-phase migration:

- 1) **Phase 1 (Hybrid Observation):** Deploy hybrid tokens (RSA+PQC) to verify performance without breaking legacy systems.
- 2) **Phase 2 (PQC-Enforcement):** Require PQC signatures for critical tools (e.g., shell execution, financial transactions) while allowing RSA for read-only tools.
- 3) **Phase 3 (Full Post-Quantum):** Retire RSA/ECDSA entirely as quantum hardware milestones are achieved.

C. Impact on Decentralized Agents and DAOs

As AI agents move toward decentralized autonomous organizations (DAOs), the immutability of audit logs becomes paramount. Our *PQC-Audit-Chain* ensures that even with the advent of large-scale quantum computers, the history of agent decisions remains cryptographically verifiable, preventing retroactive revision of automated governance actions.

D. Future Directions: Transport Security with ML-KEM

While this work focuses on authentication and integrity via ML-DSA, a complete post-quantum security posture requires protecting the confidentiality of agent communication against "Store Now, Decrypt Later" (SNDL) attacks. Future iterations of the `llm-cli` framework will explore the integration of ML-KEM (FIPS 203) to establish quantum-safe session keys for the underlying transport layer (e.g., within TLS or SSH tunnels). This would complement our signature-based approach, providing a holistic defense against both identity spoofing and unauthorized context decryption.

VIII. CONCLUSION

Securing AI agents against quantum threats is an imperative step as autonomous systems become deeply integrated into critical infrastructure. Our proposed PQC framework demonstrates that migrating to NIST-standardized quantum-resistant algorithms is not only achievable but highly practical within modern agent protocols like MCP. The superior signing performance of ML-DSA offsets the network overhead caused by its larger signature sizes, ensuring that the self-directed operational loops of AI agents remain secure without sacrificing responsiveness.

ACKNOWLEDGMENT

This research builds upon the foundational zero-trust principles implemented in the `llm-cli` open-source project.

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